

Tuesday Warm-up: Electric Fields in Matter

Prof. Jordan C. Hanson

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H	He	Li	Be	C	Ne	Na	Ar	K	Cs
0.667	0.205	24.3	5.60	1.67	0.396	24.1	1.64	43.4	59.4

Figure 1: Atomic polarizabilities, in units of $\alpha/(4\pi\epsilon_0) \times 10^{-30}$, for various atomic elements.

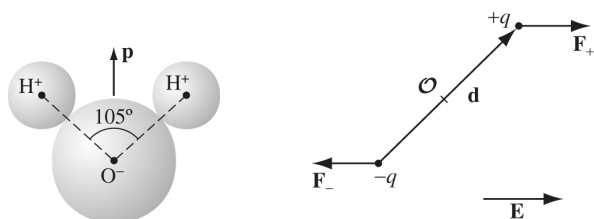


Figure 2: (Left) Water is an example of a *polar molecule*. (Right) Molecules with dipole moments experience a torque in an external field.

1 Memory Bank

- **Gradient** in spherical coordinates:

$$\nabla V = \frac{\partial V}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{\boldsymbol{\theta}} + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \hat{\boldsymbol{\phi}} \quad (1)$$

- **Product Rule 5** for use in integration by parts:

$$\nabla \cdot (f \mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f) \quad (2)$$

- **Potential of a dipole** with dipole moment $\mathbf{p}$:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2} \quad (3)$$

- **Polarizability** α of atom or molecule with induced dipole moment $\mathbf{p}$, in external field $\mathbf{E}$:

$$\mathbf{p} = \alpha \mathbf{E} \quad (4)$$

- **Torque** on a dipole with dipole moment $\mathbf{p}$ in an external field $\mathbf{E}$:

$$\mathbf{N} = \mathbf{p} \times \mathbf{E} \quad (5)$$

- Given the **polarization per unit volume**, $\mathbf{P}$:

- Bound volume charge: $\rho_b = -\nabla \cdot \mathbf{P}$
- Bound surface charge: $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$

2 Vector Calculus Technique

- Let $f = 1/r$, and $\mathbf{A} = A\hat{\mathbf{r}}$. (a) Use product rule 5 (Memory Bank) to compute $\nabla \cdot (f\mathbf{A})$. (b) Use product rule 5

(Memory Bank) to rearrange the following volume integral over a sphere of radius R :

$$\int \mathbf{A} \cdot (\nabla f) d\tau' \quad (6)$$

(c) Use the divergence theorem to switch the volume integral into a surface integral. (d) Complete the integral.

3 Polarizable Atoms and Molecules

- Suppose an atom consists of a point charge $+q$ and a uniformly charged sphere with radius a and total charge $-q$, both centered at the origin. (a) Use Gauss' law to calculate the field due to the negatively charged cloud a distance d from the origin. (b) Suppose this field strength is equal and opposite to the field pulling on $+q$. Calculate the ratio α between $p = qd$ and the field strength. (c) Show that $\alpha/(4\pi\epsilon_0)$ has units of volume.¹
- Suppose a dipole consisting of two equal and opposite charges is placed in an external field. Prove the torque on the dipole is $\mathbf{N} = \mathbf{p} \times \mathbf{E}$.
- Suppose a sphere has a uniform polarization $\mathbf{P}$. What are the (a) σ_b and (b) ρ_b ? (c) Which Legendre polynomial does the charge distribution resemble?

¹Note: the units of ϵ_0 are Farads per meter.